QP Code **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION, October 2026

Third SEMESTER

Minor Pathway Course (MPC) – MG3MPCHY200 – BIO-ORGANIC CHEMISTRY

(2025 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Answer in one sentence

Answer any **10** questions

Each question carries **1** marks

1. State the rule stating that terpenes are built from linked five-carbon units. [K] / [CO1]

2. Outline the structural difference in the number of carbon atoms between a sesquiterpene and a diterpene. [K] / [CO1]

3. Define the term essential oil. [K] / [CO1]

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4. Define the term alkaloid. [K] / [CO1]

5. Define the term substrate based on enzyme action. [K] / [CO2]

6. Explain the term reaction specificity of enzymes. [K] / [CO2]

7. Define irreversible enzyme inhibition. [K] / [CO2]

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8. Define nucleoside [K] / [CO3]

9. Name the bases present in RNA [K] / [CO3]

10. Name two fat soluble and water-soluble vitamins [K] / [CO4]

11. Name four vitamin deficiency diseases [K] / [CO4]

12. Give any two functions of Vitamin C [K] / [CO4]

[1x10 = 10]

Part B

Short answer

Answer any **4** questions

Each question carries **5** marks

13. Write a note on compound lipids. [K] / [CO1]

14. Differentiate between essential and non-essential amino acids [K] / [CO1]

15. Write short notes on apoenzyme, cofactor, coenzyme. [K] / [CO2]

16. Explain the structure of energy rich molecules. [K] / [CO3]

17. Discuss the replication of DNA. [K] / [CO3]

18. Give a brief account of vitamin deficiency diseases. [U] / [CO4]

[5x4 = 20]

Part C

Essay

Answer any **2** questions

Each question carries **10** marks

19. Write an essay on Lipids and its classification. [K] / [CO1]

20. Write an essay on mechanism of enzyme action. [U] / [CO2]

21. Write a note on role of DNA in protein biosynthesis. [U] / [CO3]

22. Write an essay on water soluble vitamins. [K] / [CO4]

[10x2 = 20]